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Benchtop Power Station

This memo shares an update on an eight-year effort to enhance our ESET/MXET labs instruments for DC power measurement. Progress is fantastic in most measures except for faculty awareness and support. We cover the origin, purpose and updated strategy (2026 fall) for the Power Station project by David Malawey.

Right now, it is important to gain faculty's attention so the effort becomes supported, and students gain the intended benefits. It's also crucial to communicate the unmet needs for capstone projects in our department, so the Lab Staff and professors can have aligned understandings.

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History:

This project is approximately 8 years of effort, being summarized for new readers.

- Data Gathering – 2018
- Start of Design – 2023
- Workbench Established – 2024
- Used by Students – 2025
- Project Update - 2026

In the following diagram, find the introductory diagram for the power bench, created in 2023. This diagram and further documentation was published on Github at github.com/MXET/resources, and the next stage is to simplify the power bench and rebuild it in smaller segments.

The station's purpose is to make instrumentation available which supports our modern capstone design efforts.

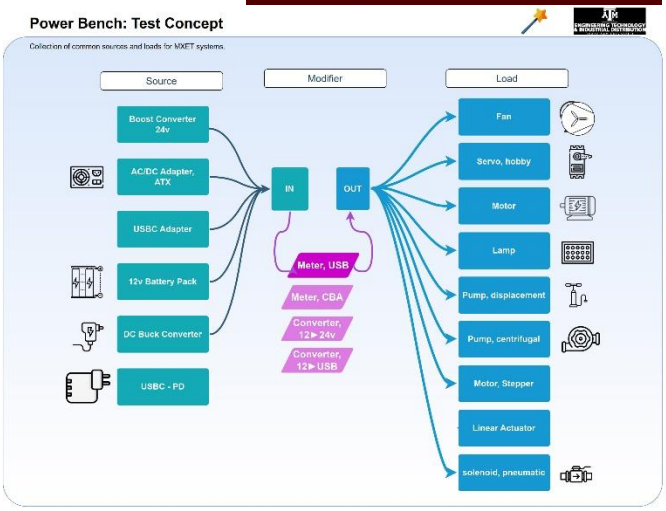


Figure 1 Instrument Utility Diagram

The above diagram comes from the documentation of the Power Station's first implementation. From left to right, this diagram indicates a user can select a power source, an instrument, and a power load. It is a flexible station for electronics power evaluation.

Problem 1: Measuring

Over 50% of our capstone teams create embedded devices without suitable equipment. Our traditional instruments, i.e. benchtop power supply (PSU), benchtop oscilloscope, and benchtop multimeters are a slow and inefficient solution. They consume excessive workspace and sometimes lack relevant functions for battery-powered device development.

In Capstone 2016 to present, students build projects with embedded power circuits. The native power sources include batteries and DC adapters, and the circuits contain onboard power conditioning. However, our PIC (Product Innovation Cellar) electronics lab contains generic, traditional instruments which are often unsuitable for embedded device development.

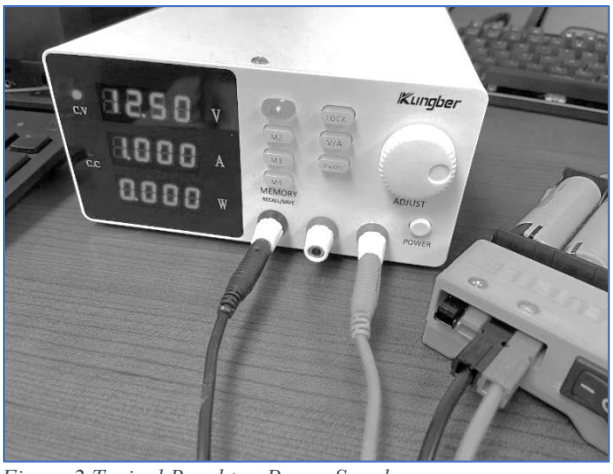


Figure 2 Typical Benchtop Power Supply

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The problem in our present approach is bulky machines, incompatible cables, and a high effort for each team to prepare tests amid development. Further, we lack of automatic logging. Traditional benchtop machines are great for some tasks and poor for many other tasks at-hand.

Above, find an example DC power supply representing our main apparatus for power measurement. In lab, this also becomes the apparatus for other tasks discussed later.

Problem 2: Wiring

As a Lab Coordinator, it is easy to identify a pattern that slows the capstone designers' progress. To connect the embedded devices to the legacy instruments, we find chains of adapters at each student station. An example chain of conductors:

- (power supply) banana plug to
- alligator clip to
- 1-core copper wire to
- solderless breadboard to
- Dupont cable to
- pins of embedded device
- (embedded device has USB connector)

Below, find an example benchtop testing setup from a 2019 MXET capstone team. Note that 90% of the space is taken by the instruments and adapting wires while 10% is taken by the student project. This should be inverted, ideally.

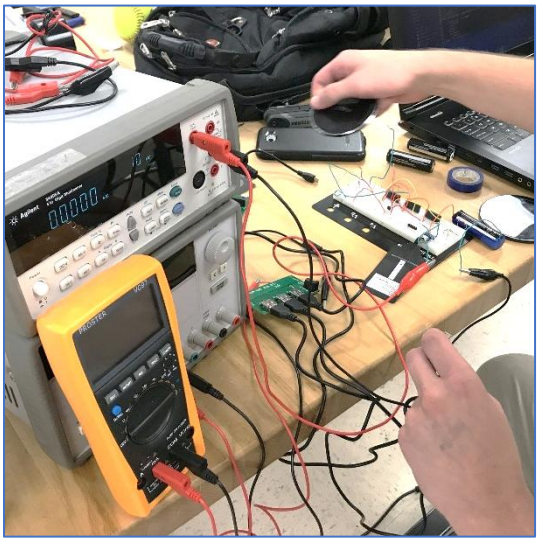


Figure 3 Students wire up a power test

These makeshift solutions expand the working space, complicate the lab inventory, and lock students in a lending-pattern where the good connectors are always in use by another team. Alternately, students order per-project adapters and build temporary or unreliable connections, repeatedly and uniquely for each team.

Results:

- More parts and PO's than needed
- Surprise shipping delays for teams
- Lab cables returned in piles
- 90% preparation and 10% design

If we accept this level of lab work to be normal and productive, it is likely based on an outdated experience of electronics labs.

Problem 3: Embedded Device Behavior

This section describes two nuances of our embedded devices: The unique power-draw behavior, and the power profile's dependency on software (or firmware).

The **ETID Department's core strength lies in embedded systems**, focusing on compact, low-power applications with intricate circuitry that distinguish our work from traditional electrical engineering disciplines. Cultivating this core competency is essential, as it directly drives our reputation and serves as a major draw for employers hiring our graduates.

Embedded Device Behavior:

In Capstone, students frequently test embedded systems which have a unique power behavior. Teams use benchtop power supplies as power meters, by recording the amperage and voltage from the display. They make manual measurements to suit simple goals such as:

- Selecting an appropriate battery
- Validating function of a freshly soldered component
- Power-up a transceiver for a coms test

The test subjects are lightweight boards driven by systems-on-chip now including bluetooth, Wi-Fi, and long-range radio transceivers. Frequently, they operate at low power levels with millisecond-range pulses of power demand, where measurements require:

- fine time-domain sampling
- mA level resolution

Please note: the basic project goals do NOT require fine accuracy but without it, teams are misinformed coarsely. It is the combination of device behavior and instrument behavior that frequently hinders teams.

Below, find a typical 5-second sample of the power for an IoT device, measured in milliamps. **The energy consumption is confined to momentary pulses at sporadic intervals**. This makes it challenging to discover the average power, peak power, and total energy consumed. This profile is unsuitable for measuring with a benchtop power supply. Still, most students are not aware and they become misled by their own power readings.

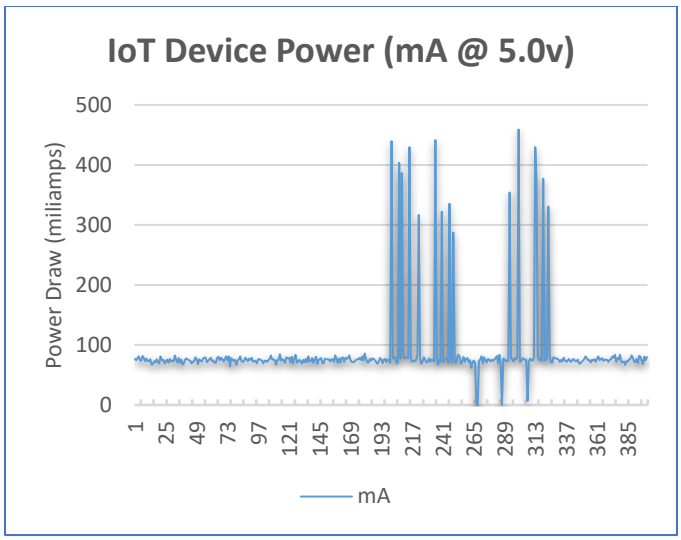


Figure 4 Power Draw of an IoT device

It's desired to provide the students with a better option than our general benchtop instruments for many of their recurring tasks. To recap and summarize this problem: Students face two goals

1) knowing peak power draw to size DC adapters
2) knowing the average power to meet battery-life targets

For these two metrics, it is evident that student teams are hindered. A variety of new instruments are now available that better match embedded devices.

Software-defined Circuits

Another recuring factor expands the need to measure power, or at least observe it. Unlike the past stationary project designs, our modern embedded projects exhibit power demand depending on software (firmware, in fact), which is concurrently designed with the circuit. Whether or not students are aware, system power will change with each line of code that is uploaded to the controller.

The student teams encounter changing power needs and adjust their work in the lab. While weekly meetings with faculty focus on functionality, their workspace is adapted to keep trials powered up. A power measurement is reported as one session, but in the lab power trials occur continually. This trend is driven by firmware changes and by all the components that respond to firmware changes. In the table, see how the teams handle devices from the early stages to late stages of design.

Component	Early Trials	Later Trials
Motor	Activating	Setting duty cycles
Servo	No-load movement	Loaded movement
WiFi transceiver	Connectivity	Data transmission
Software	Sample code	Custom routines
Whole System	Power by PSU	Power by Battery

This is only a small sample of the many changes that occur as the students develop a design and each one changes the power demand behavior of the system. Actuators in the projects are often impossible to test at an early stage. For example, popular stepper motors can be rated at 30 watts, then driven with sample code at 100 watts, and with some adjustments fall to 10 watts. Students cannot interpret complicated datasheets and cannot run an accurate trial until they discover smooth settings which are discovered in the final weeks of the semester.

In both solid-state projects and heavily actuated projects, power observations must be made continually as students build an idea of how they work. **The teams see a simple solution: keep the instruments at the project table for the whole semester.**

Since power tests (or just powered tests) must run continually, all our bulky devices and wiring nests reside with the project full-time. For the capstone labs, the work surfaces become fully consumed mid-semester and students stack electronics on each other to continue. More space may help but the better solution is modernized equipment.

Task: Current Measurement

The most accessible solution for evaluating embedded electronics is a simple USB meter – a device now abundant and economical. The advantage is that these devices are specifically designed for light loads and fine resolution. The drawback is a marketplace of low-quality devices, unbranded models and missing datasheets.

Between 2018-2022, I evaluated inexpensive online-sourced power meters and narrowed down a few with accurate useful measurements, and which carried brand names with traceable model numbers.

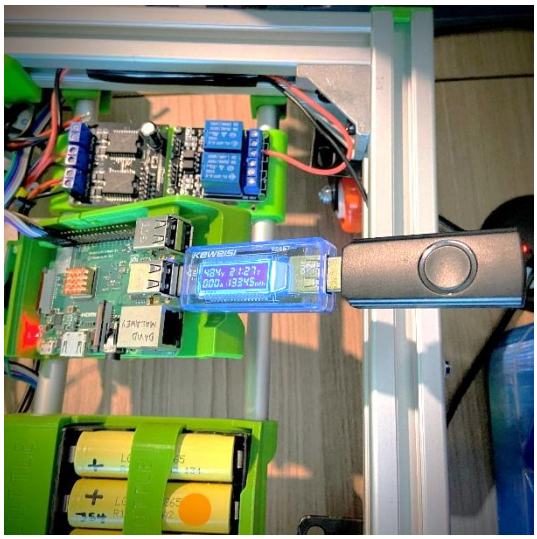


Figure 5 Simple effective USB meter

I validated the devices experimentally, then researched online for calibration data. Other engineers have blogged outlines of the detailed behavior, so these cheap meters can be trusted for engineering projects, for the main metrics of interest.

So instead of perfect datasheets, we have engineer-authored instructions which describe both operation and pitfalls. These instructions are perfectly suitable if they are reviewed by a

trained engineer. We have a qualified engineer operating as a lab coordinator for this purpose.

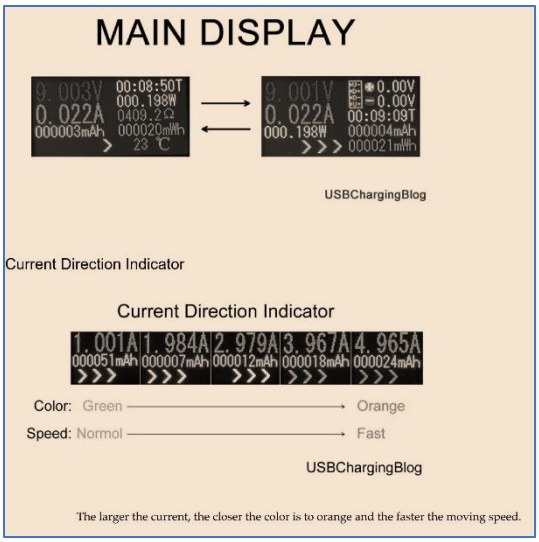


Figure 6 – snippet from Engineer's instruction for USB meter

Can a cheap $10 instrument help our lab? My study says YES, many USB instruments are suitable and accurate. The devices should be selected and set up with engineering specificity, which becomes the focus of the power bench design. Secondly, the key functions and issues should be summarized before handoff to the undergrad audience. Simplification is required.

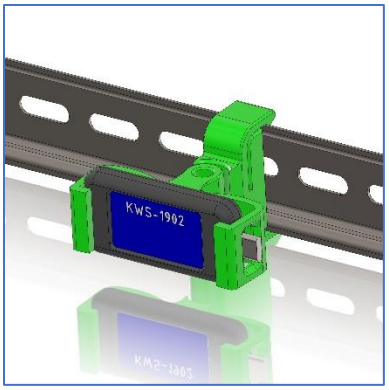
Simplified Testing

To accommodate this need for simplification, I took steps both in hardware setup and in the documentation. For a focal subset of tasks, the instrument has pre-made instructions and pre-made wiring connections.

Mechanical and electrical integration also simplifies the user experience. As shown below, a custom bracket mounts the USB meter into the testing station (and this bracket readily adjusts if a new instrument is adopted in the lab.) The goal is to integrate the device electrically and visually into the benchtop station to communicate its place in a testing process.

Instead of requiring students to configure all the parts of a power evaluation, they simply bring their device as an input or an output device. Refer to the first figure having green and blue blocks. A student can simply connect their source (green block) or their load (blue block) to begin a trial.

The Benchtop Power station is an idealized design and it cannot b



Task: DC Loading

One unmet need in 2018 was: **applying a calibrated DC load to a circuit** and logging power metrics. In the below image, find the CBA V (Computerized Battery Analyzer) product from West Mountain Radio, sourced at Mouser Electronics.

This compact, and accurate device proved easy to use, with simple lightweight software, compatible with any PC. It was the perfect educational instrument for student projects with rapid results. **We have no other solution which can load a battery or power circuit with a specified wattage or amperage, and log data** over time.

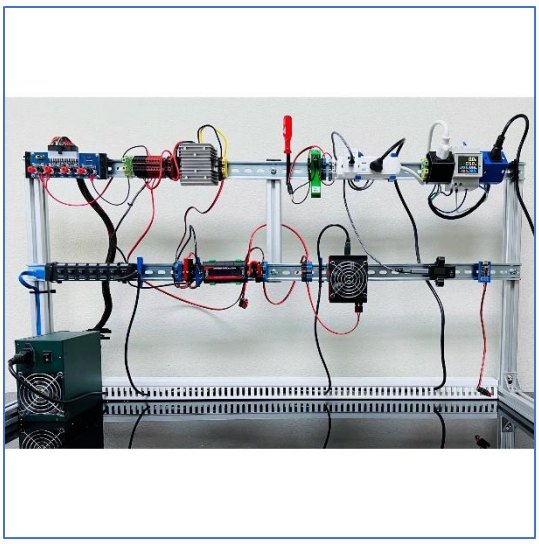
The CBA unit was validated by the Lab Coordinator during the development of the SCUTTLE Robot, and HEBI robot setup in MXET labs (2018, 2019). It proved useful and versatile and deserved a carryover to the capstone lab for students to access it.



Figure 7 Programmable Power Load & Logger

The CBA unit received a mounting solution and is now integration-ready for any DIN-rail equipped workstation. Below, find an image of the mounted unit.





Task: Simpler Wiring

Regarding complex wiring described in the "Problem2" section, USB adapters are part of the solution. In our present-day senior classes, up to 50% of the electronics connections begin or end with a USB connection. For those connections, the manual red+black individual power setups add several cables or adapters and make desktop measurements much more confusing.

For supporting this range of project needs, it is advantageous to add a number of USB connectors and cables to our instrumentation shelves, and to include USB terminals in our power station. But the details for these advisable USB devices will be held for another discussion for the brevity of this memo.

First Implementation:

Draft 1 of the Power Bench is shown in the next image. It is not aesthetic, having a 0-dollar budget. **Without instruction, it is difficult to navigate**. When this was created, the plan was to work TOGETHER with students. Attempts to arrange info sessions became formalized into "workshops" or "class sessions" and the administrative planning halted as the professors had no background information to prioritize the effort.

The power bench draft build had a simple purpose of making the instruments visible for students.

Administrative Blockage.

In 2023, the goal of sharing this apparatus with students was interrupted as I was reporting to the MXET PC (Program Coordinator), newly assigned to an entirely new faculty PC. This PC would need multiple years to understand the style of our capstone program and the needs of the students.

My approach to brief the new PC was a complete failure, as I requested multiple meetings, and the PC was unavailable except for 5-minute exchanges.

The next best option was to **document extensively and permit the faculty to read at their leisure**, which did not go as planned. Our professor in charge was educated in mechanical engineering and lacked the electronics fundamental education to interpret the documentation in terms of relevance to our student projects.

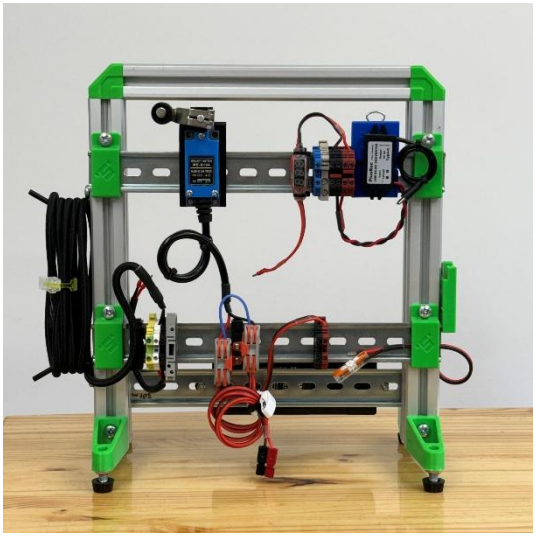
New Direction (2026)

The plan for fall 2026 is to disassemble the benchtop station into smaller stations, and simplify the first station for reintroduction. The station will be smaller, have only one instrument, and fewer "moving parts." The choice of instrument is the CBA unit because students successfully operated this without instruction, in 2025-2026.

The broader options and goals can be grouped into smaller stations and set up, to give time for students and faculty to observe the utility and form communication to suggest adjustments.

Visually, we will make the testing "stations" more intuitive, and that will mean sacrificing some options and utility in the short term. Connectors may be reduced to fewer options and become compatible with a smaller range of devices. Then, the input and

output connections can be made fixed and arranged in an orderly way.



The footprint for a station will be reduced, similar to the benchtop frame shown above (a DC12V source station). The example photo shows a DC sourcing station that simply plugs into a 120v wall outlet and provides 12v power at multiple output connections. It is reconfigurable with only a Philips1 screwdriver. The utility is straightforward, and the output power can be quickly switched on and off with a toggle switch.

New Options are Ready

The design elements for this Power Bench project consist of several engineering efforts that are completed and organized by David Malawey for the ETID Labs. They include:

- bracket designs that fit electronics modules onto DIN rail
- fastener studies for hardware best suited to merge DIN rail, 3030 extrusions, and electronics cables.
- Parametric designs with equation-based geometry which readily adapt for custom circuits.
- Electronics circuit designs, and selected modules for switches and terminals, 3D CAD modeled and integration-ready in custom systems.

Based on the completed designs, our updated stations will be more robust and aesthetic. Future laboratory equipment, instrument stations, and custom workspaces can be rapidly assembled with modular documentation that equips learners and Teaching Assistants with minimal effort.

The photos and benchtop stations depicted in this publication are only a small portion of the ready designed elements as of 2026.

Feedback

If our educators have any feedback for this activity, this memo should be referenced to establish context. Also, the Power Bench documentation may be used. The faculty are encouraged to read the “Power Bench Queries and Solutions” page which is listed in the references.

References:

1) The sample dataset for power measurement of IoT device is an approximation reflecting source data in the following journal article:

Fine Time Measurement for the Internet of Things: A Practical Approach Using ESP32. Valentin Vales et al. IEEE Internet of Things Journal Vol 9. Oct.1, 2022

2) Expanded documentation for the Power Station published by David Malawey at the following github repository as of 2026.10:

https://github.com/MXET/resources

3) The resources including the Power Station and student engineering support documents, from the above github page are viewable as a webpage:

https://qr.net/mxet

4) Example Queries that show experimental options, available in PDF form at:

https://github.com/MXET/resources/blob/52e25d3daa446509823e69ba1ae855bc5f3c8c2d/doc/2024_PowerBench_Examples.pdf